

Expanded Grade 5 Math Practice

Eight Units | 240 Original Questions | Worked Solutions and Answer Key

This workbook follows the eight-unit scope supplied for California Math Expressions Grade 5, Volumes 1 and 2. Questions and explanations in this book are newly written practice material; none are copied from the textbook.

Directions: Show your work. Use drawings, equations, tables, or models where they help. Check answers in the worked-solutions section only after attempting each problem.

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Volume 1

Chapters 1-4: Whole-number operations, measurement, and geometry.

Chapter 1: Multiplication and Division Word Problems

Concept: Multiplicative comparisons and word problems

What it means: Multiplication is useful when every group has the same amount or when one amount is a certain number of times another amount. **Example:** If 4 boxes hold 3 books each, draw 4 equal groups of 3. Then write $4 \times 3 = 12$. **Remember:** The words *each*, *groups of*, and *times as many* often signal multiplication.

Visual model



Questions

1. A library packs 4 boxes with 11 books in each box. How many books are packed?
2. A library packs 5 boxes with 15 books in each box. How many books are packed?
3. A library packs 6 boxes with 19 books in each box. How many books are packed?
4. A library packs 7 boxes with 23 books in each box. How many books are packed?
5. A library packs 8 boxes with 27 books in each box. How many books are packed?
6. A library packs 9 boxes with 31 books in each box. How many books are packed?
7. A library packs 10 boxes with 35 books in each box. How many books are packed?
8. A library packs 11 boxes with 39 books in each box. How many books are packed?
9. A library packs 12 boxes with 43 books in each box. How many books are packed?
10. A library packs 13 boxes with 47 books in each box. How many books are packed?

11. Maya collected 66 shells. Diego collected 4 times as many. How many shells did Diego collect?
12. Maya collected 72 shells. Diego collected 5 times as many. How many shells did Diego collect?
13. Maya collected 78 shells. Diego collected 6 times as many. How many shells did Diego collect?
14. Maya collected 84 shells. Diego collected 7 times as many. How many shells did Diego collect?
15. Maya collected 90 shells. Diego collected 8 times as many. How many shells did Diego collect?

Concept: Expressions, parentheses, and properties

What it means: An expression is a math sentence without an equals sign. Parentheses act like a fence: solve what is inside first. **Example:** In $5 \times (8 + 2)$, add $8 + 2$ first to make 10, then multiply 5×10 . The distributive property gives another path: $5 \times 8 + 5 \times 2$. **Remember:** Both paths give the same total.

Visual model

$$5 \times (8 + 2) = 5 \times 8 + 5 \times 2$$

Questions

16. Evaluate $(20 + 48) \times 7$.
17. Evaluate $(21 + 51) \times 8$.
18. Evaluate $(22 + 54) \times 9$.
19. Evaluate $(23 + 57) \times 10$.
20. Evaluate $(24 + 60) \times 11$.
21. Use the distributive property to find 11×23 by breaking 23 into $21 + 2$.
22. Use the distributive property to find 12×24 by breaking 24 into $22 + 2$.

23. Use the distributive property to find 13×25 by breaking 25 into $23 + 2$.
24. Use the distributive property to find 14×26 by breaking 26 into $24 + 2$.
25. Use the distributive property to find 15×27 by breaking 27 into $25 + 2$.

Concept: Factors

What it means: Factors are the numbers multiplied together to make a product. **Example:** The factor pairs for 24 are 1×24 , 2×12 , 3×8 , and 4×6 . Every number in those pairs is a factor of 24. **Remember:** Start with 1 and work upward until the pairs repeat.

Visual model

$$24 = 1 \times 24 = 2 \times 12 = 3 \times 8 = 4 \times 6$$

Questions

26. List every factor of 78.
27. List every factor of 81.
28. List every factor of 84.
29. List every factor of 87.
30. List every factor of 90.

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

1.1 Answer: 44

Multiply the equal groups: $4 \times 11 = 44$.

1.2 Answer: 75

Multiply the equal groups: $5 \times 15 = 75$.

1.3 Answer: 114

Multiply the equal groups: $6 \times 19 = 114$.

1.4 Answer: 161

Multiply the equal groups: $7 \times 23 = 161$.

1.5 Answer: 216

Multiply the equal groups: $8 \times 27 = 216$.

1.6 Answer: 279

Multiply the equal groups: $9 \times 31 = 279$.

1.7 Answer: 350

Multiply the equal groups: $10 \times 35 = 350$.

1.8 Answer: 429

Multiply the equal groups: $11 \times 39 = 429$.

1.9 Answer: 516

Multiply the equal groups: $12 \times 43 = 516$.

1.10 Answer: 611

Multiply the equal groups: $13 \times 47 = 611$.

1.11 Answer: 264

A multiplicative comparison uses multiplication: $66 \times 4 = 264$.

1.12 Answer: 360

A multiplicative comparison uses multiplication: $72 \times 5 = 360$.

1.13 Answer: 468

A multiplicative comparison uses multiplication: $78 \times 6 = 468$.

1.14 Answer: 588

A multiplicative comparison uses multiplication: $84 \times 7 = 588$.

1.15 Answer: 720

A multiplicative comparison uses multiplication: $90 \times 8 = 720$.

1.16 Answer: 476

Work inside parentheses first: $20 + 48 = 68$. Then multiply: $68 \times 7 = 476$.

1.17 Answer: 576

Work inside parentheses first: $21 + 51 = 72$. Then multiply: $72 \times 8 = 576$.

1.18 Answer: 684

Work inside parentheses first: $22 + 54 = 76$. Then multiply: $76 \times 9 = 684$.

1.19 Answer: 800

Work inside parentheses first: $23 + 57 = 80$. Then multiply: $80 \times 10 = 800$.

1.20 Answer: 924

Work inside parentheses first: $24 + 60 = 84$. Then multiply: $84 \times 11 = 924$.

1.21 Answer: 253

$11 \times (21 + 2) = (11 \times 21) + (11 \times 2) = 231 + 22 = 253$.

1.22 Answer: 288

$12 \times (22 + 2) = (12 \times 22) + (12 \times 2) = 264 + 24 = 288$.

1.23 Answer: 325

$13 \times (23 + 2) = (13 \times 23) + (13 \times 2) = 299 + 26 = 325$.

1.24 Answer: 364

$14 \times (24 + 2) = (14 \times 24) + (14 \times 2) = 336 + 28 = 364$.

1.25 Answer: 405

$15 \times (25 + 2) = (15 \times 25) + (15 \times 2) = 375 + 30 = 405.$

1.26 Answer: 1, 2, 3, 6, 13, 26, 39, 78

Pair factors that multiply to 78: 1×78 , 2×39 , 3×26 , 6×13 .

1.27 Answer: 1, 3, 9, 27, 81

Pair factors that multiply to 81: 1×81 , 3×27 , 9×9 .

1.28 Answer: 1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84

Pair factors that multiply to 84: 1×84 , 2×42 , 3×28 , 4×21 , 6×14 , 7×12 .

1.29 Answer: 1, 3, 29, 87

Pair factors that multiply to 87: 1×87 , 3×29 .

1.30 Answer: 1, 2, 3, 5, 6, 9, 10, 15, 18, 30, 45, 90

Pair factors that multiply to 90: 1×90 , 2×45 , 3×30 , 5×18 , 6×15 , 9×10 .

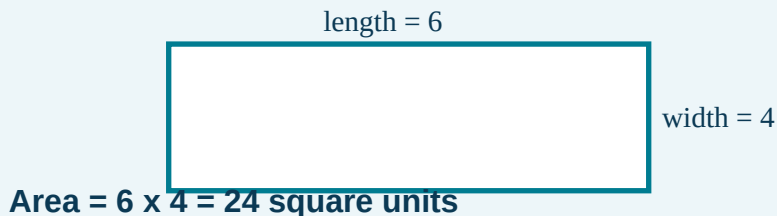
Chapter 2: Perimeter and Area

Concept: Perimeter and area of rectangles

What it means: Perimeter is the distance all the way around a shape. Area is the amount of flat space inside it.

Example: A 6-by-4 rectangle has perimeter $6 + 4 + 6 + 4 = 20$ units, but area $6 \times 4 = 24$ square units. **Remember:** Perimeter uses regular units; area uses square units.

Visual model



Questions

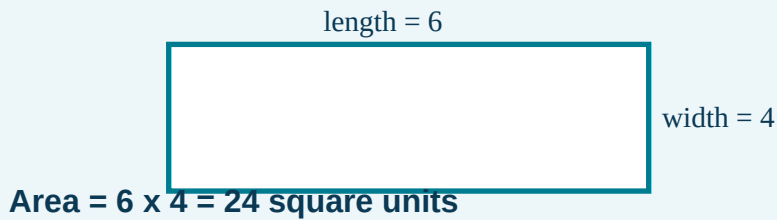
1. A rectangle is 6 cm long and 3 cm wide. Find its perimeter and area.
2. A rectangle is 7 cm long and 4 cm wide. Find its perimeter and area.
3. A rectangle is 8 cm long and 5 cm wide. Find its perimeter and area.
4. A rectangle is 9 cm long and 6 cm wide. Find its perimeter and area.
5. A rectangle is 10 cm long and 7 cm wide. Find its perimeter and area.
6. A rectangle is 11 cm long and 8 cm wide. Find its perimeter and area.
7. A rectangle is 12 cm long and 9 cm wide. Find its perimeter and area.
8. A rectangle is 13 cm long and 10 cm wide. Find its perimeter and area.
9. A rectangle is 14 cm long and 11 cm wide. Find its perimeter and area.
10. A rectangle is 15 cm long and 12 cm wide. Find its perimeter and area.

Concept: Finding an unknown side length

What it means: A rectangle's area is length $\times$ width. If you know the area and one side, you can use division to find the missing side. **Example:** An area of 35 square units with a width of 5 units means 35 divided by 5 = 7 units for

the length. **Remember:** Multiply your two side lengths to check the area.

Visual model



Questions

11. A garden has an area of 126 square meters and a width of 7 meters. What is its length?
12. A garden has an area of 152 square meters and a width of 8 meters. What is its length?
13. A garden has an area of 180 square meters and a width of 9 meters. What is its length?
14. A garden has an area of 210 square meters and a width of 10 meters. What is its length?
15. A garden has an area of 242 square meters and a width of 11 meters. What is its length?
16. A garden has an area of 276 square meters and a width of 12 meters. What is its length?
17. A garden has an area of 312 square meters and a width of 13 meters. What is its length?
18. A garden has an area of 350 square meters and a width of 14 meters. What is its length?
19. A garden has an area of 390 square meters and a width of 15 meters. What is its length?
20. A garden has an area of 432 square meters and a width of 16 meters. What is its length?

Concept: Area models for multiplication

What it means: An area model splits a harder multiplication problem into smaller, easier rectangles. **Example:** For 23×4 , split 23 into 20 and 3. Find $20 \times 4 = 80$ and $3 \times 4 = 12$; then add $80 + 12 = 92$. **Remember:** Every part of the split must be included in the final sum.

Visual model



Questions

21. Use an area model or the distributive property to find 111×7 .
22. Use an area model or the distributive property to find 124×8 .
23. Use an area model or the distributive property to find 137×9 .
24. Use an area model or the distributive property to find 141×10 .
25. Use an area model or the distributive property to find 154×11 .
26. Use an area model or the distributive property to find 167×12 .
27. Use an area model or the distributive property to find 171×13 .
28. Use an area model or the distributive property to find 184×14 .
29. Use an area model or the distributive property to find 197×15 .
30. Use an area model or the distributive property to find 201×16 .

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

2.1 Answer: **Perimeter = 18 cm; area = 18 sq cm**

Perimeter = $2 \times (6 + 3) = 18$ cm. Area = $6 \times 3 = 18$ square centimeters.

2.2 Answer: **Perimeter = 22 cm; area = 28 sq cm**

Perimeter = $2 \times (7 + 4) = 22$ cm. Area = $7 \times 4 = 28$ square centimeters.

2.3 Answer: **Perimeter = 26 cm; area = 40 sq cm**

Perimeter = $2 \times (8 + 5) = 26$ cm. Area = $8 \times 5 = 40$ square centimeters.

2.4 Answer: Perimeter = 30 cm; area = 54 sq cm

Perimeter = $2 \times (9 + 6) = 30$ cm. Area = $9 \times 6 = 54$ square centimeters.

2.5 Answer: Perimeter = 34 cm; area = 70 sq cm

Perimeter = $2 \times (10 + 7) = 34$ cm. Area = $10 \times 7 = 70$ square centimeters.

2.6 Answer: Perimeter = 38 cm; area = 88 sq cm

Perimeter = $2 \times (11 + 8) = 38$ cm. Area = $11 \times 8 = 88$ square centimeters.

2.7 Answer: Perimeter = 42 cm; area = 108 sq cm

Perimeter = $2 \times (12 + 9) = 42$ cm. Area = $12 \times 9 = 108$ square centimeters.

2.8 Answer: Perimeter = 46 cm; area = 130 sq cm

Perimeter = $2 \times (13 + 10) = 46$ cm. Area = $13 \times 10 = 130$ square centimeters.

2.9 Answer: Perimeter = 50 cm; area = 154 sq cm

Perimeter = $2 \times (14 + 11) = 50$ cm. Area = $14 \times 11 = 154$ square centimeters.

2.10 Answer: Perimeter = 54 cm; area = 180 sq cm

Perimeter = $2 \times (15 + 12) = 54$ cm. Area = $15 \times 12 = 180$ square centimeters.

2.11 Answer: 18 m

Area = length $\times$ width, so length = 126 divided by $7 = 18$ meters.

2.12 Answer: 19 m

Area = length $\times$ width, so length = 152 divided by $8 = 19$ meters.

2.13 Answer: 20 m

Area = length $\times$ width, so length = 180 divided by $9 = 20$ meters.

2.14 Answer: 21 m

Area = length $\times$ width, so length = 210 divided by $10 = 21$ meters.

2.15 Answer: 22 m

Area = length $\times$ width, so length = 242 divided by $11 = 22$ meters.

2.16 Answer: 23 m

Area = length $\times$ width, so length = 276 divided by $12 = 23$ meters.

2.17 Answer: 24 m

Area = length $\times$ width, so length = 312 divided by $13 = 24$ meters.

2.18 Answer: 25 m

Area = length $\times$ width, so length = 350 divided by $14 = 25$ meters.

2.19 Answer: 26 m

Area = length $\times$ width, so length = 390 divided by $15 = 26$ meters.

2.20 Answer: 27 m

Area = length $\times$ width, so length = 432 divided by $16 = 27$ meters.

2.21 Answer: 777

Break 111 into $110 + 1$: $(110 \times 7) + (1 \times 7) = 770 + 7 = 777$.

2.22 Answer: 992

Break 124 into $120 + 4$: $(120 \times 8) + (4 \times 8) = 960 + 32 = 992$.

2.23 Answer: 1233

Break 137 into $130 + 7$: $(130 \times 9) + (7 \times 9) = 1170 + 63 = 1233$.

2.24 Answer: 1410

Break 141 into $140 + 1$: $(140 \times 10) + (1 \times 10) = 1400 + 10 = 1410$.

2.25 Answer: 1694

Break 154 into $150 + 4$: $(150 \times 11) + (4 \times 11) = 1650 + 44 = 1694$.

2.26 Answer: 2004

Break 167 into $160 + 7$: $(160 \times 12) + (7 \times 12) = 1920 + 84 = 2004$.

2.27 Answer: 2223

Break 171 into $170 + 1$: $(170 \times 13) + (1 \times 13) = 2210 + 13 = 2223$.

2.28 Answer: 2576

Break 184 into $180 + 4$: $(180 \times 14) + (4 \times 14) = 2520 + 56 = 2576$.

2.29 Answer: 2955

Break 197 into $190 + 7$: $(190 \times 15) + (7 \times 15) = 2850 + 105 = 2955$.

2.30 Answer: 3216

Break 201 into $200 + 1$: $(200 \times 16) + (1 \times 16) = 3200 + 16 = 3216$.

Chapter 3: Addition and Subtraction of Whole Numbers and Decimals

Concept: Adding and subtracting whole numbers

What it means: Each digit has a place value, such as ones, tens, hundreds, or thousands. **Example:** Put ones under ones, tens under tens, and so on. Regroup 10 ones as 1 ten when adding, or trade 1 ten for 10 ones when subtracting. **Remember:** Neat vertical columns protect the place values.

Visual model

Thousands	Hundreds	Tens	Ones

Line up each digit by its place value

Questions

1. Find $27,321 + 13,109$.
2. Find $31,642 + 15,218$.
3. Find $35,963 + 17,327$.
4. Find $40,284 + 19,436$.
5. Find $44,605 + 21,545$.
6. Find $48,926 + 23,654$.
7. Find $53,247 + 25,763$.
8. Find $57,568 + 27,872$.
9. Find $101,853 - 38,569$.
10. Find $103,170 - 39,410$.
11. Find $104,487 - 40,251$.
12. Find $105,804 - 41,092$.

13. Find $107,121 - 41,933$.

14. Find $108,438 - 42,774$.

15. Find $109,755 - 43,615$.

Concept: Adding and subtracting decimals

What it means: Decimal places also have place values: tenths, hundredths, and thousandths. **Example:** For $3.40 + 0.65$, line up the decimal points. Add hundredths, then tenths, then ones to get 4.05. **Remember:** Add zeros as placeholders, but never move a decimal point.

Visual model

$$\begin{array}{r} 2.40 \\ \times 3.50 \\ \hline 8.40 \end{array}$$

Two decimal places in each factor

Questions

16. Find $19.47 + 2.18$.

17. Find $20.47 + 2.28$.

18. Find $21.47 + 2.38$.

19. Find $22.47 + 2.48$.

20. Find $23.47 + 2.58$.

21. Find $24.47 + 2.68$.

22. Find $25.47 + 2.78$.

23. Find $26.47 + 2.88$.

24. Find $32.36 - 3.87$.

25. Find $33.36 - 3.97$.

26. Find $34.36 - 4.07$.

27. Find $35.36 - 4.17$.

28. Find $36.36 - 4.27$.

29. Find $37.36 - 4.37$.

30. Find $38.36 - 4.47$.

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

3.1 Answer: 40430

Align place values and add: $27,321 + 13,109 = 40,430$.

3.2 Answer: 46860

Align place values and add: $31,642 + 15,218 = 46,860$.

3.3 Answer: 53290

Align place values and add: $35,963 + 17,327 = 53,290$.

3.4 Answer: 59720

Align place values and add: $40,284 + 19,436 = 59,720$.

3.5 Answer: 66150

Align place values and add: $44,605 + 21,545 = 66,150$.

3.6 Answer: 72580

Align place values and add: $48,926 + 23,654 = 72,580$.

3.7 Answer: 79010

Align place values and add: $53,247 + 25,763 = 79,010$.

3.8 Answer: 85440

Align place values and add: $57,568 + 27,872 = 85,440$.

3.9 Answer: 63284

Align place values and subtract: $101,853 - 38,569 = 63,284$.

3.10 Answer: 63760

Align place values and subtract: $103,170 - 39,410 = 63,760$.

3.11 Answer: 64236

Align place values and subtract: $104,487 - 40,251 = 64,236$.

3.12 Answer: 64712

Align place values and subtract: $105,804 - 41,092 = 64,712$.

3.13 Answer: 65188

Align place values and subtract: $107,121 - 41,933 = 65,188$.

3.14 Answer: 65664

Align place values and subtract: $108,438 - 42,774 = 65,664$.

3.15 Answer: 66140

Align place values and subtract: $109,755 - 43,615 = 66,140$.

3.16 Answer: 21.65

Line up the decimal points: $19.47 + 2.18 = 21.65$.

3.17 Answer: 22.75

Line up the decimal points: $20.47 + 2.28 = 22.75$.

3.18 Answer: 23.85

Line up the decimal points: $21.47 + 2.38 = 23.85$.

3.19 Answer: 24.95

Line up the decimal points: $22.47 + 2.48 = 24.95$.

3.20 Answer: 26.05

Line up the decimal points: $23.47 + 2.58 = 26.05$.

3.21 Answer: 27.15

Line up the decimal points: $24.47 + 2.68 = 27.15$.

3.22 Answer: 28.25

Line up the decimal points: $25.47 + 2.78 = 28.25$.

3.23 Answer: 29.35

Line up the decimal points: $26.47 + 2.88 = 29.35$.

3.24 Answer: 28.49

Line up the decimal points: $32.36 - 3.87 = 28.49$.

3.25 Answer: 29.39

Line up the decimal points: $33.36 - 3.97 = 29.39$.

3.26 Answer: 30.29

Line up the decimal points: $34.36 - 4.07 = 30.29$.

3.27 Answer: 31.19

Line up the decimal points: $35.36 - 4.17 = 31.19$.

3.28 Answer: 32.09

Line up the decimal points: $36.36 - 4.27 = 32.09$.

3.29 Answer: 32.99

Line up the decimal points: $37.36 - 4.37 = 32.99$.

3.30 Answer: 33.89

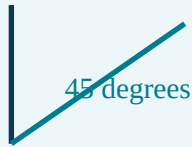
Line up the decimal points: $38.36 - 4.47 = 33.89$.

Chapter 4: Circles, Angles, and Polygons

Concept: Measuring and classifying angles

What it means: An angle is made by two rays meeting at a vertex. The size is the turn between them, measured in degrees. **Example:** A 45-degree angle is acute, a 90-degree angle is right, a 120-degree angle is obtuse, and a 180-degree angle is straight. **Remember:** Compare the opening, not the length of the rays.

Visual model



acute < 90

right $= 90$

Questions

1. Classify an angle that measures 25 degrees.
2. Classify an angle that measures 48 degrees.
3. Classify an angle that measures 72 degrees.
4. Classify an angle that measures 90 degrees.
5. Classify an angle that measures 105 degrees.
6. Classify an angle that measures 128 degrees.
7. Classify an angle that measures 160 degrees.
8. Classify an angle that measures 180 degrees.
9. Classify an angle that measures 37 degrees.
10. Classify an angle that measures 145 degrees.

Concept: Angles on a straight line

What it means: A straight line is a half-turn, so its angle measure is 180 degrees. **Example:** If one part of a straight line is 70 degrees, the missing part is $180 - 70 = 110$ degrees. **Remember:** Add both adjacent angles to check for

180 degrees.

Visual model



$$70 + 110 = 180 \text{ degrees}$$

Questions

11. Two angles form a straight line. One measures 97 degrees. What is the other angle?
12. Two angles form a straight line. One measures 104 degrees. What is the other angle?
13. Two angles form a straight line. One measures 111 degrees. What is the other angle?
14. Two angles form a straight line. One measures 118 degrees. What is the other angle?
15. Two angles form a straight line. One measures 125 degrees. What is the other angle?
16. Two angles form a straight line. One measures 132 degrees. What is the other angle?
17. Two angles form a straight line. One measures 139 degrees. What is the other angle?
18. Two angles form a straight line. One measures 146 degrees. What is the other angle?
19. Two angles form a straight line. One measures 153 degrees. What is the other angle?
20. Two angles form a straight line. One measures 160 degrees. What is the other angle?

Concept: Polygons

What it means: A polygon is a closed, flat shape made only from straight line segments. **Example:** A pentagon has 5 sides and 5 vertices. A hexagon has 6 sides and 6 vertices. **Remember:** Curved sides or an open gap mean the shape is not a polygon.

Visual model



5 sides = pentagon

Questions

21. How many sides and vertices does a triangle have?
22. How many sides and vertices does a quadrilateral have?
23. How many sides and vertices does a pentagon have?
24. How many sides and vertices does a hexagon have?
25. How many sides and vertices does a heptagon have?
26. How many sides and vertices does a octagon have?
27. How many sides and vertices does a nonagon have?
28. How many sides and vertices does a decagon have?
29. How many sides and vertices does a square have?
30. How many sides and vertices does a equilateral triangle have?

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

4.1 Answer: **acute**

25 degrees is acute because it is less than 90 degrees.

4.2 Answer: **acute**

48 degrees is acute because it is less than 90 degrees.

4.3 Answer: **acute**

72 degrees is acute because it is less than 90 degrees.

4.4 Answer: right

90 degrees is right because it is equal to 90 degrees.

4.5 Answer: obtuse

105 degrees is obtuse because it is between 90 and 180 degrees.

4.6 Answer: obtuse

128 degrees is obtuse because it is between 90 and 180 degrees.

4.7 Answer: obtuse

160 degrees is obtuse because it is between 90 and 180 degrees.

4.8 Answer: straight

180 degrees is straight because it is equal to 180 degrees.

4.9 Answer: acute

37 degrees is acute because it is less than 90 degrees.

4.10 Answer: obtuse

145 degrees is obtuse because it is between 90 and 180 degrees.

4.11 Answer: 83 degrees

Angles on a straight line total 180 degrees: $180 - 97 = 83$ degrees.

4.12 Answer: 76 degrees

Angles on a straight line total 180 degrees: $180 - 104 = 76$ degrees.

4.13 Answer: 69 degrees

Angles on a straight line total 180 degrees: $180 - 111 = 69$ degrees.

4.14 Answer: 62 degrees

Angles on a straight line total 180 degrees: $180 - 118 = 62$ degrees.

4.15 Answer: 55 degrees

Angles on a straight line total 180 degrees: $180 - 125 = 55$ degrees.

4.16 Answer: 48 degrees

Angles on a straight line total 180 degrees: $180 - 132 = 48$ degrees.

4.17 Answer: 41 degrees

Angles on a straight line total 180 degrees: $180 - 139 = 41$ degrees.

4.18 Answer: 34 degrees

Angles on a straight line total 180 degrees: $180 - 146 = 34$ degrees.

4.19 Answer: 27 degrees

Angles on a straight line total 180 degrees: $180 - 153 = 27$ degrees.

4.20 Answer: 20 degrees

Angles on a straight line total 180 degrees: $180 - 160 = 20$ degrees.

4.21 Answer: 3 sides and 3 vertices

A triangle has 3 sides. Each corner is a vertex, so it has 3 vertices.

4.22 Answer: 4 sides and 4 vertices

A quadrilateral has 4 sides. Each corner is a vertex, so it has 4 vertices.

4.23 Answer: 5 sides and 5 vertices

A pentagon has 5 sides. Each corner is a vertex, so it has 5 vertices.

4.24 Answer: 6 sides and 6 vertices

A hexagon has 6 sides. Each corner is a vertex, so it has 6 vertices.

4.25 Answer: 7 sides and 7 vertices

A heptagon has 7 sides. Each corner is a vertex, so it has 7 vertices.

4.26 Answer: 8 sides and 8 vertices

A octagon has 8 sides. Each corner is a vertex, so it has 8 vertices.

4.27 Answer: 9 sides and 9 vertices

A nonagon has 9 sides. Each corner is a vertex, so it has 9 vertices.

4.28 Answer: 10 sides and 10 vertices

A decagon has 10 sides. Each corner is a vertex, so it has 10 vertices.

4.29 Answer: 4 sides and 4 vertices

A square has 4 sides. Each corner is a vertex, so it has 4 vertices.

4.30 Answer: 3 sides and 3 vertices

A equilateral triangle has 3 sides. Each corner is a vertex, so it has 3 vertices.

Volume 2

Chapters 5-8: Fractions, volume, coordinates, and advanced operations.

Chapter 5: Addition and Subtraction with Fractions

Concept: Equivalent fractions

What it means: Equivalent fractions look different but name the same amount. **Example:** One-half and two-fourths both cover half of a shape. Multiply the numerator and denominator of $\frac{1}{2}$ by 2 to make $\frac{2}{4}$. **Remember:** Multiply or divide the top and bottom by the same number.

Visual model



$\frac{1}{2}$



$\frac{2}{4}$

same amount

Questions

1. Write an equivalent fraction for $\frac{1}{2}$ with denominator 4.
2. Write an equivalent fraction for $\frac{2}{3}$ with denominator 6.
3. Write an equivalent fraction for $\frac{3}{4}$ with denominator 8.
4. Write an equivalent fraction for $\frac{3}{5}$ with denominator 10.
5. Write an equivalent fraction for $\frac{4}{7}$ with denominator 14.
6. Write an equivalent fraction for $\frac{5}{8}$ with denominator 16.
7. Write an equivalent fraction for $\frac{2}{9}$ with denominator 18.
8. Write an equivalent fraction for $\frac{3}{10}$ with denominator 20.

Concept: Adding fractions

What it means: You can only add fractions when the pieces are the same size. The denominator names the piece size. **Example:** For $\frac{1}{3} + \frac{1}{6}$, change $\frac{1}{3}$ into $\frac{2}{6}$. Then add $\frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$. **Remember:** Find a common denominator before adding the numerators.

Visual model



$\frac{1}{2}$



$\frac{2}{4}$

same amount

Questions

9. Find $\frac{1}{3} + \frac{1}{6}$.

10. Find $\frac{2}{5} + \frac{1}{10}$.

11. Find $\frac{3}{4} + \frac{1}{8}$.

12. Find $\frac{2}{3} + \frac{1}{9}$.

13. Find $\frac{3}{5} + \frac{1}{4}$.

14. Find $\frac{5}{6} + \frac{1}{3}$.

15. Find $\frac{7}{8} + \frac{1}{4}$.

Concept: Subtracting fractions

What it means: Fraction subtraction also needs equal-size pieces. **Example:** For $\frac{3}{4} - \frac{1}{6}$, use twelfths: $\frac{3}{4} = \frac{9}{12}$ and $\frac{1}{6} = \frac{2}{12}$. Then $\frac{9}{12} - \frac{2}{12} = \frac{7}{12}$. **Remember:** Subtract only the numerators after the denominators match.

Visual model



$\frac{1}{2}$



$\frac{2}{4}$

same amount

Questions

16. Find $\frac{4}{7} - \frac{2}{7}$.

17. Find $\frac{5}{9} - \frac{2}{3}$.

18. Find $\frac{7}{10} - \frac{3}{5}$.

19. Find $\frac{11}{12} - \frac{1}{6}$.

20. Find $\frac{3}{4} - \frac{2}{5}$.

21. Find $\frac{5}{6} - \frac{1}{2}$.

22. Find $\frac{7}{9} - \frac{1}{3}$.

Concept: Comparing fractions and decimals

What it means: To compare two numbers, write them in the same form or use a benchmark such as 0, $\frac{1}{2}$, or 1.

Example: $\frac{3}{4}$ equals 0.75, so $\frac{3}{4}$ is greater than 0.70. **Remember:** The symbol $>$ points toward the smaller number, like an open mouth.

Visual model

$$\begin{array}{r} 2.40 \\ \times 3.50 \\ \hline 8.40 \end{array}$$

Two decimal places in each factor

Questions

23. Compare $\frac{3}{4}$ and 0.58. Use $<$, $>$, or $=$.

24. Compare $\frac{5}{8}$ and 0.70. Use $<$, $>$, or $=$.

25. Compare $\frac{7}{12}$ and 0.60. Use $<$, $>$, or $=$.

26. Compare $\frac{9}{10}$ and 0.90. Use $<$, $>$, or $=$.

27. Compare $\frac{4}{9}$ and 0.42. Use $<$, $>$, or $=$.

28. Compare $\frac{3}{5}$ and 0.55. Use $<$, $>$, or $=$.

29. Compare $\frac{7}{8}$ and 0.80. Use $<$, $>$, or $=$.

30. Compare $\frac{2}{5}$ and 0.45. Use $<$, $>$, or $=$.

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

5.1 Answer: $\frac{2}{4}$

Multiply numerator and denominator by 2: $\frac{1}{2} = \frac{2}{4}$.

5.2 Answer: $\frac{4}{6}$

Multiply numerator and denominator by 2: $\frac{2}{3} = \frac{4}{6}$.

5.3 Answer: $\frac{6}{8}$

Multiply numerator and denominator by 2: $\frac{3}{4} = \frac{6}{8}$.

5.4 Answer: $\frac{6}{10}$

Multiply numerator and denominator by 2: $\frac{3}{5} = \frac{6}{10}$.

5.5 Answer: $\frac{8}{14}$

Multiply numerator and denominator by 2: $\frac{4}{7} = \frac{8}{14}$.

5.6 Answer: $\frac{10}{16}$

Multiply numerator and denominator by 2: $\frac{5}{8} = \frac{10}{16}$.

5.7 Answer: $\frac{4}{18}$

Multiply numerator and denominator by 2: $\frac{2}{9} = \frac{4}{18}$.

5.8 Answer: $\frac{6}{20}$

Multiply numerator and denominator by 2: $\frac{3}{10} = \frac{6}{20}$.

5.9 Answer: $\frac{1}{2}$

Use a common denominator, then add. The sum simplifies to $\frac{1}{2}$.

5.10 Answer: $\frac{1}{2}$

Use a common denominator, then add. The sum simplifies to $\frac{1}{2}$.

5.11 Answer: $\frac{7}{8}$

Use a common denominator, then add. The sum simplifies to $\frac{7}{8}$.

5.12 Answer: $\frac{7}{9}$

Use a common denominator, then add. The sum simplifies to $\frac{7}{9}$.

5.13 Answer: $\frac{17}{20}$

Use a common denominator, then add. The sum simplifies to $\frac{17}{20}$.

5.14 Answer: $1\frac{1}{6}$

Use a common denominator, then add. The sum simplifies to $1\frac{1}{6}$.

5.15 Answer: $1\frac{1}{8}$

Use a common denominator, then add. The sum simplifies to $1\frac{1}{8}$.

5.16 Answer: $\frac{2}{7}$

Use a common denominator, then subtract. The difference simplifies to $\frac{2}{7}$.

5.17 Answer: $-1\frac{8}{9}$

Use a common denominator, then subtract. The difference simplifies to $-1\frac{8}{9}$.

5.18 Answer: $\frac{1}{10}$

Use a common denominator, then subtract. The difference simplifies to $\frac{1}{10}$.

5.19 Answer: $\frac{3}{4}$

Use a common denominator, then subtract. The difference simplifies to $\frac{3}{4}$.

5.20 Answer: $\frac{7}{20}$

Use a common denominator, then subtract. The difference simplifies to $\frac{7}{20}$.

5.21 Answer: $\frac{1}{3}$

Use a common denominator, then subtract. The difference simplifies to $\frac{1}{3}$.

5.22 Answer: $\frac{4}{9}$

Use a common denominator, then subtract. The difference simplifies to $\frac{4}{9}$.

5.23 Answer: $>$

Convert $\frac{3}{4}$ to a decimal or use a common benchmark. $0.75 > 0.58$.

5.24 Answer: $<$

Convert $\frac{5}{8}$ to a decimal or use a common benchmark. $0.625 < 0.70$.

5.25 Answer: $<$

Convert $\frac{7}{12}$ to a decimal or use a common benchmark. $0.583 < 0.60$.

5.26 Answer: $<$

Convert $\frac{9}{10}$ to a decimal or use a common benchmark. $0.9 < 0.90$.

5.27 Answer: $>$

Convert $\frac{4}{9}$ to a decimal or use a common benchmark. $0.444 > 0.42$.

5.28 Answer: $>$

Convert $\frac{3}{5}$ to a decimal or use a common benchmark. $0.6 > 0.55$.

5.29 Answer: $>$

Convert $\frac{7}{8}$ to a decimal or use a common benchmark. $0.875 > 0.80$.

5.30 Answer: $<$

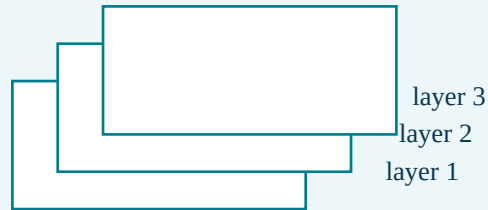
Convert $\frac{2}{5}$ to a decimal or use a common benchmark. $0.4 < 0.45$.

Chapter 6: Volume, Capacity, and Weight

Concept: Volume of rectangular prisms

What it means: Volume counts the cubic units that fill a solid box. **Example:** A prism 4 units long, 3 units wide, and 2 units high contains $4 \times 3 \times 2 = 24$ cubic units. **Remember:** Volume has three measurements and uses cubic units.

Visual model



12 cubes in each layer x 3 layers = 36 cubic units

Questions

1. Find the volume of a rectangular prism with length 3 cm, width 4 cm, and height 2 cm.
2. Find the volume of a rectangular prism with length 4 cm, width 5 cm, and height 3 cm.
3. Find the volume of a rectangular prism with length 5 cm, width 6 cm, and height 4 cm.
4. Find the volume of a rectangular prism with length 6 cm, width 7 cm, and height 5 cm.
5. Find the volume of a rectangular prism with length 7 cm, width 8 cm, and height 6 cm.
6. Find the volume of a rectangular prism with length 8 cm, width 9 cm, and height 7 cm.
7. Find the volume of a rectangular prism with length 9 cm, width 10 cm, and height 8 cm.
8. Find the volume of a rectangular prism with length 10 cm, width 11 cm, and height 9 cm.
9. Find the volume of a rectangular prism with length 11 cm, width 12 cm, and height 10 cm.
10. Find the volume of a rectangular prism with length 12 cm, width 13 cm, and height 11 cm.

Concept: Capacity, mass, and length conversions

What it means: A conversion changes the unit but not the amount. **Example:** Since 1 foot = 12 inches, 3 feet = $3 \times 12 = 36$ inches. **Remember:** Moving from a bigger unit to a smaller unit makes the number larger.

Visual model

$$3 \text{ feet} \quad \underline{\hspace{1cm}} \times 12 \quad \underline{\hspace{1cm}} \quad 36 \text{ inches}$$

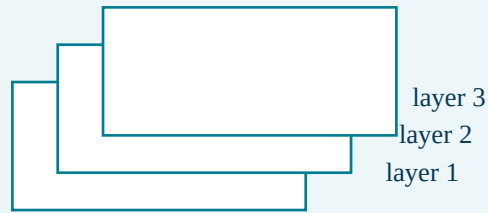
Questions

11. Convert 3 feet to inches.
12. Convert 5 yards to feet.
13. Convert 4 pints to cups.
14. Convert 3 quarts to pints.
15. Convert 2 gallons to quarts.
16. Convert 6 pounds to ounces.
17. Convert 7 kilograms to grams.
18. Convert 9 meters to centimeters.
19. Convert 4 liters to milliliters.
20. Convert 8 tons to pounds.

Concept: Volume as equal layers

What it means: A rectangular prism is built from equal layers of cubes. **Example:** If one layer has 12 cubes and the box has 3 layers, it has $12 \times 3 = 36$ cubic units. **Remember:** Count a layer, then multiply by the number of layers.

Visual model



12 cubes in each layer x 3 layers = 36 cubic units

Questions

21. A box has 7 equal layers of unit cubes. Each layer has 25 cubes. What is the volume?
22. A box has 8 equal layers of unit cubes. Each layer has 26 cubes. What is the volume?
23. A box has 9 equal layers of unit cubes. Each layer has 27 cubes. What is the volume?
24. A box has 10 equal layers of unit cubes. Each layer has 28 cubes. What is the volume?
25. A box has 11 equal layers of unit cubes. Each layer has 29 cubes. What is the volume?
26. A box has 12 equal layers of unit cubes. Each layer has 30 cubes. What is the volume?
27. A box has 13 equal layers of unit cubes. Each layer has 31 cubes. What is the volume?
28. A box has 14 equal layers of unit cubes. Each layer has 32 cubes. What is the volume?
29. A box has 15 equal layers of unit cubes. Each layer has 33 cubes. What is the volume?
30. A box has 16 equal layers of unit cubes. Each layer has 34 cubes. What is the volume?

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

6.1 Answer: **24 cubic cm**

Volume = $l \times w \times h = 3 \times 4 \times 2 = 24$ cubic centimeters.

6.2 Answer: **60 cubic cm**

Volume = $l \times w \times h = 4 \times 5 \times 3 = 60$ cubic centimeters.

6.3 Answer: **120 cubic cm**

Volume = $l \times w \times h = 5 \times 6 \times 4 = 120$ cubic centimeters.

6.4 Answer: 210 cubic cm

Volume = $l \times w \times h = 6 \times 7 \times 5 = 210$ cubic centimeters.

6.5 Answer: 336 cubic cm

Volume = $l \times w \times h = 7 \times 8 \times 6 = 336$ cubic centimeters.

6.6 Answer: 504 cubic cm

Volume = $l \times w \times h = 8 \times 9 \times 7 = 504$ cubic centimeters.

6.7 Answer: 720 cubic cm

Volume = $l \times w \times h = 9 \times 10 \times 8 = 720$ cubic centimeters.

6.8 Answer: 990 cubic cm

Volume = $l \times w \times h = 10 \times 11 \times 9 = 990$ cubic centimeters.

6.9 Answer: 1320 cubic cm

Volume = $l \times w \times h = 11 \times 12 \times 10 = 1320$ cubic centimeters.

6.10 Answer: 1716 cubic cm

Volume = $l \times w \times h = 12 \times 13 \times 11 = 1716$ cubic centimeters.

6.11 Answer: 36 inches

Use the conversion relationship: 3 feet = 36 inches.

6.12 Answer: 15 feet

Use the conversion relationship: 5 yards = 15 feet.

6.13 Answer: 8 cups

Use the conversion relationship: 4 pints = 8 cups.

6.14 Answer: 6 pints

Use the conversion relationship: 3 quarts = 6 pints.

6.15 Answer: 8 quarts

Use the conversion relationship: 2 gallons = 8 quarts.

6.16 Answer: 96 ounces

Use the conversion relationship: 6 pounds = 96 ounces.

6.17 Answer: 7000 grams

Use the conversion relationship: 7 kilograms = 7000 grams.

6.18 Answer: 900 centimeters

Use the conversion relationship: 9 meters = 900 centimeters.

6.19 Answer: 4000 milliliters

Use the conversion relationship: 4 liters = 4000 milliliters.

6.20 Answer: 16000 pounds

Use the conversion relationship: 8 tons = 16000 pounds.

6.21 Answer: 175 cubic units

Multiply cubes in one layer by layers: $25 \times 7 = 175$ cubic units.

6.22 Answer: 208 cubic units

Multiply cubes in one layer by layers: $26 \times 8 = 208$ cubic units.

6.23 Answer: 243 cubic units

Multiply cubes in one layer by layers: $27 \times 9 = 243$ cubic units.

6.24 Answer: 280 cubic units

Multiply cubes in one layer by layers: $28 \times 10 = 280$ cubic units.

6.25 Answer: 319 cubic units

Multiply cubes in one layer by layers: $29 \times 11 = 319$ cubic units.

6.26 Answer: 360 cubic units

Multiply cubes in one layer by layers: $30 \times 12 = 360$ cubic units.

6.27 Answer: 403 cubic units

Multiply cubes in one layer by layers: $31 \times 13 = 403$ cubic units.

6.28 Answer: 448 cubic units

Multiply cubes in one layer by layers: $32 \times 14 = 448$ cubic units.

6.29 Answer: 495 cubic units

Multiply cubes in one layer by layers: $33 \times 15 = 495$ cubic units.

6.30 Answer: 544 cubic units

Multiply cubes in one layer by layers: $34 \times 16 = 544$ cubic units.

Chapter 7: Integers and the Coordinate Plane

Concept: Integers and absolute value

What it means: Integers include positive numbers, negative numbers, and zero. Absolute value is how far a number is from zero on a number line. **Example:** -5 is five steps left of zero, so $|-5| = 5$. **Remember:** Absolute value is never negative.

Visual model

$|-5| = 5$: five steps from zero



Questions

1. Find the absolute value of -15.
2. Find the absolute value of 8.
3. Find the absolute value of -3.
4. Find the absolute value of 0.
5. Find the absolute value of 22.
6. Find the absolute value of -11.
7. Find the absolute value of 14.
8. Find the absolute value of -9.
9. Find the absolute value of 6.
10. Find the absolute value of -20.
11. Find the absolute value of 17.
12. Find the absolute value of -4.

13. Find the absolute value of 25.

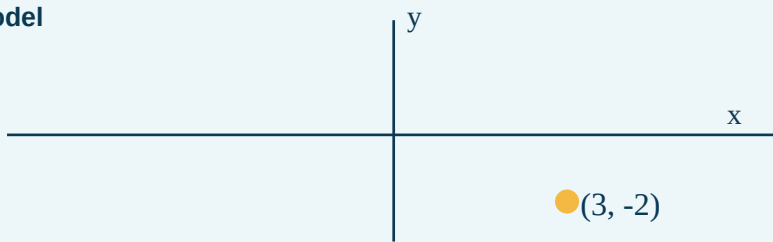
14. Find the absolute value of -7.

15. Find the absolute value of 12.

Concept: The coordinate plane

What it means: A coordinate plane uses a horizontal x-axis and vertical y-axis. An ordered pair is read x first, then y. **Example:** To plot (3, -2), move 3 right and then 2 down. **Remember:** Start at the origin (0, 0) every time.

Visual model



Questions

16. In which quadrant is the point (-2, 4)?

17. In which quadrant is the point (3, 5)?

18. In which quadrant is the point (-4, -1)?

19. In which quadrant is the point (6, -3)?

20. In which quadrant is the point (-5, 2)?

21. In which quadrant is the point (1, -6)?

22. In which quadrant is the point (-3, -4)?

23. In which quadrant is the point (5, 1)?

24. In which quadrant is the point (-1, 7)?

25. In which quadrant is the point (4, -5)?
26. In which quadrant is the point (-6, 3)?
27. In which quadrant is the point (2, -2)?
28. In which quadrant is the point (-7, -3)?
29. In which quadrant is the point (8, 4)?
30. In which quadrant is the point (-2, -8)?

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

7.1 Answer: 15

Absolute value is distance from 0, so $|-15| = 15$.

7.2 Answer: 8

Absolute value is distance from 0, so $|8| = 8$.

7.3 Answer: 3

Absolute value is distance from 0, so $|-3| = 3$.

7.4 Answer: 0

Absolute value is distance from 0, so $|0| = 0$.

7.5 Answer: 22

Absolute value is distance from 0, so $|22| = 22$.

7.6 Answer: 11

Absolute value is distance from 0, so $|-11| = 11$.

7.7 Answer: 14

Absolute value is distance from 0, so $|14| = 14$.

7.8 Answer: 9

Absolute value is distance from 0, so $|-9| = 9$.

7.9 Answer: 6

Absolute value is distance from 0, so $|6| = 6$.

7.10 Answer: 20

Absolute value is distance from 0, so $|-20| = 20$.

7.11 Answer: 17

Absolute value is distance from 0, so $|17| = 17$.

7.12 Answer: 4

Absolute value is distance from 0, so $|-4| = 4$.

7.13 Answer: 25

Absolute value is distance from 0, so $|25| = 25$.

7.14 Answer: 7

Absolute value is distance from 0, so $|-7| = 7$.

7.15 Answer: 12

Absolute value is distance from 0, so $|12| = 12$.

7.16 Answer: Quadrant II

The x-coordinate is negative and the y-coordinate is positive, so the point is in Quadrant II.

7.17 Answer: Quadrant I

The x-coordinate is positive and the y-coordinate is positive, so the point is in Quadrant I.

7.18 Answer: Quadrant III

The x-coordinate is negative and the y-coordinate is negative, so the point is in Quadrant III.

7.19 Answer: Quadrant IV

The x-coordinate is positive and the y-coordinate is negative, so the point is in Quadrant IV.

7.20 Answer: Quadrant II

The x-coordinate is negative and the y-coordinate is positive, so the point is in Quadrant II.

7.21 Answer: Quadrant IV

The x-coordinate is positive and the y-coordinate is negative, so the point is in Quadrant IV.

7.22 Answer: Quadrant III

The x-coordinate is negative and the y-coordinate is negative, so the point is in Quadrant III.

7.23 Answer: Quadrant I

The x-coordinate is positive and the y-coordinate is positive, so the point is in Quadrant I.

7.24 Answer: Quadrant II

The x-coordinate is negative and the y-coordinate is positive, so the point is in Quadrant II.

7.25 Answer: Quadrant IV

The x-coordinate is positive and the y-coordinate is negative, so the point is in Quadrant IV.

7.26 Answer: Quadrant II

The x-coordinate is negative and the y-coordinate is positive, so the point is in Quadrant II.

7.27 Answer: Quadrant IV

The x-coordinate is positive and the y-coordinate is negative, so the point is in Quadrant IV.

7.28 Answer: Quadrant III

The x-coordinate is negative and the y-coordinate is negative, so the point is in Quadrant III.

7.29 Answer: Quadrant I

The x-coordinate is positive and the y-coordinate is positive, so the point is in Quadrant I.

7.30 Answer: Quadrant III

The x-coordinate is negative and the y-coordinate is negative, so the point is in Quadrant III.

Chapter 8: Advanced Multiplication and Division with Fractions and Decimals

Concept: Multiplying fractions by whole numbers

What it means: Multiplying a fraction by a whole number means taking that fraction several times. **Example:** $\frac{3}{4} \times 4$ means four groups of $\frac{3}{4}$. Multiply to get $\frac{12}{4} = 3$. **Remember:** Simplify after multiplying, and turn an improper fraction into a mixed number if needed.

Visual model

Thousands	Hundreds	Tens	Ones

Line up each digit by its place value

Questions

1. Find $\frac{1}{3} \times 6$.
2. Find $\frac{2}{5} \times 10$.
3. Find $\frac{3}{4} \times 8$.
4. Find $\frac{5}{6} \times 12$.
5. Find $\frac{3}{8} \times 16$.
6. Find $\frac{7}{10} \times 20$.
7. Find $\frac{4}{9} \times 18$.
8. Find $\frac{5}{12} \times 24$.

Concept: Dividing fractions by whole numbers

What it means: Dividing a fraction by a whole number splits that fraction into equal groups. **Example:** $\frac{3}{4}$ divided by 2 means split three fourths between 2 groups: $\frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$. **Remember:** Divide by a whole number by multiplying by its reciprocal, 1 over that number.

Visual model

Thousands	Hundreds	Tens	Ones

Line up each digit by its place value

Questions

9. Find $\frac{3}{4}$ divided by 2.
10. Find $\frac{5}{6}$ divided by 3.
11. Find $\frac{7}{8}$ divided by 4.
12. Find $\frac{2}{3}$ divided by 5.
13. Find $\frac{9}{10}$ divided by 3.
14. Find $\frac{5}{12}$ divided by 2.
15. Find $\frac{7}{9}$ divided by 7.

Concept: Long division and remainders

What it means: Long division helps share a large amount into equal groups. **Example:** In 864 divided by 7, divide, multiply, subtract, and bring down at each place. The remainder must be smaller than 7. **Remember:** Check with divisor $\times$ quotient + remainder = dividend.

Visual model

$$7 \overline{) 864} = 123 \text{ R } 3$$

$$7 \times 123 + 3 = 864$$

Questions

16. Divide 864 by 7. Give a quotient and remainder.

17. Divide 975 by 8. Give a quotient and remainder.
18. Divide 1,234 by 9. Give a quotient and remainder.
19. Divide 1,465 by 6. Give a quotient and remainder.
20. Divide 2,387 by 5. Give a quotient and remainder.
21. Divide 3,210 by 4. Give a quotient and remainder.
22. Divide 1,756 by 3. Give a quotient and remainder.
23. Divide 2,689 by 7. Give a quotient and remainder.

Concept: Decimal multiplication and division

What it means: Decimal operations use place value just like whole-number operations. **Example:** For 2.4×3.5 , multiply $24 \times 35 = 840$, then place two decimal digits to get 8.40. **Remember:** Estimate first so you can tell whether the decimal point makes sense.

Visual model

$$\begin{array}{r} 2.40 \\ \times 3.50 \\ \hline 8.40 \end{array}$$

Two decimal places in each factor

Questions

24. Find 2.40×3.50 .
25. Find 1.25×4.00 .
26. Find 3.60 divided by 0.50.
27. Find 7.20 divided by 0.30.
28. Find 4.08×2.00 .

29. Find 6.25 divided by 0.50.

30. Find 8.40×1.20 .

Chapter Answers and Explained Solutions

Use these explanations to check both the answer and the concept used to find it.

8.1 Answer: 2

Multiply the whole number: $\frac{1}{3} \times 6 = 2$.

8.2 Answer: 4

Multiply the whole number: $\frac{2}{5} \times 10 = 4$.

8.3 Answer: 6

Multiply the whole number: $\frac{3}{4} \times 8 = 6$.

8.4 Answer: 10

Multiply the whole number: $\frac{5}{6} \times 12 = 10$.

8.5 Answer: 6

Multiply the whole number: $\frac{3}{8} \times 16 = 6$.

8.6 Answer: 14

Multiply the whole number: $\frac{7}{10} \times 20 = 14$.

8.7 Answer: 8

Multiply the whole number: $\frac{4}{9} \times 18 = 8$.

8.8 Answer: 10

Multiply the whole number: $\frac{5}{12} \times 24 = 10$.

8.9 Answer: $\frac{3}{8}$

Dividing by 2 means multiplying by $\frac{1}{2}$: $\frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$.

8.10 Answer: $\frac{5}{18}$

Dividing by 3 means multiplying by $\frac{1}{3}$: $\frac{5}{6} \times \frac{1}{3} = \frac{5}{18}$.

8.11 Answer: $\frac{7}{32}$

Dividing by 4 means multiplying by $\frac{1}{4}$: $\frac{7}{8} \times \frac{1}{4} = \frac{7}{32}$.

8.12 Answer: $\frac{2}{15}$

Dividing by 5 means multiplying by $\frac{1}{5}$: $\frac{2}{3} \times \frac{1}{5} = \frac{2}{15}$.

8.13 Answer: $\frac{3}{10}$

Dividing by 3 means multiplying by $\frac{1}{3}$: $\frac{9}{10} \times \frac{1}{3} = \frac{3}{10}$.

8.14 Answer: $\frac{5}{24}$

Dividing by 2 means multiplying by $\frac{1}{2}$: $\frac{5}{12} \times \frac{1}{2} = \frac{5}{24}$.

8.15 Answer: $\frac{1}{9}$

Dividing by 7 means multiplying by $\frac{1}{7}$: $\frac{7}{9} \times \frac{1}{7} = \frac{1}{9}$.

8.16 Answer: 123 R 3

864 divided by 7 = 123 remainder 3, because $7 \times 123 + 3 = 864$.

8.17 Answer: 121 R 7

975 divided by 8 = 121 remainder 7, because $8 \times 121 + 7 = 975$.

8.18 Answer: 137 R 1

1,234 divided by 9 = 137 remainder 1, because $9 \times 137 + 1 = 1,234$.

8.19 Answer: 244 R 1

1,465 divided by 6 = 244 remainder 1, because $6 \times 244 + 1 = 1,465$.

8.20 Answer: 477 R 2

2,387 divided by 5 = 477 remainder 2, because $5 \times 477 + 2 = 2,387$.

8.21 Answer: 802 R 2

3,210 divided by 4 = 802 remainder 2, because $4 \times 802 + 2 = 3,210$.

8.22 Answer: 585 R 1

1,756 divided by 3 = 585 remainder 1, because $3 \times 585 + 1 = 1,756$.

8.23 Answer: 384 R 1

2,689 divided by 7 = 384 remainder 1, because $7 \times 384 + 1 = 2,689$.

8.24 Answer: 8.40

Compute the decimal operation: $2.40 \times 3.50 = 8.40$.

8.25 Answer: 5.00

Compute the decimal operation: $1.25 \times 4.00 = 5.00$.

8.26 Answer: 7.20

Compute the decimal operation: $3.60 \text{ divided by } 0.50 = 7.20$.

8.27 Answer: 24.00

Compute the decimal operation: $7.20 \text{ divided by } 0.30 = 24.00$.

8.28 Answer: 8.16

Compute the decimal operation: $4.08 \times 2.00 = 8.16$.

8.29 Answer: 12.50

Compute the decimal operation: $6.25 \text{ divided by } 0.50 = 12.50$.

8.30 Answer: 10.08

Compute the decimal operation: $8.40 \times 1.20 = 10.08$.